



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
06-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

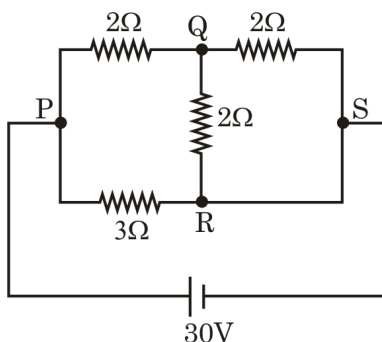
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. For the circuit shown in figure, if source of voltage is ideal, then :-



(A) Current through PR is 10A.

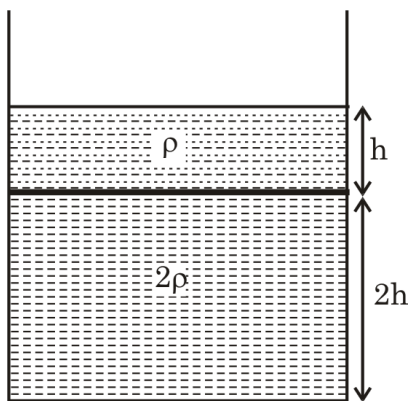
(B) Current through RS is 15 A.

(C) Current through RS is 12 A

(D) Potential difference $V_{PQ} = \frac{2}{3} V_{PR}$.

Space for Rough Work

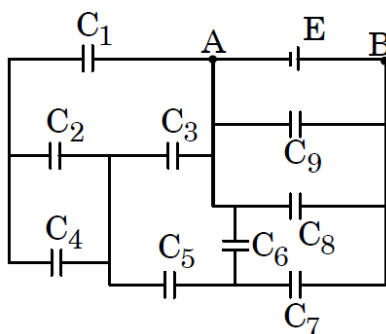
2. The is a specially designed vernier calipers in which 4 main scale division are equal to 10 vernier scale division. In the zero error, 5th and 10th division of vernier scales coincides with 1mm and 3mm mark on scale respectively and zero of the vernier scale is left to the zero of the main scale. During taking measurement 2nd and 7th divisions of the vernier scale coincide with 23mm and 25 mm mark on main scale respectively.
(1 main scale division = 1mm)
- (A) Least count of this vernier calipers is 0.6 mm.
(B) Instrument has negative zero error of 1mm.
(C) Measurement of the length is 22.2 mm.
(D) Measurement of the length is 23.2 mm.
3. A fixed cylindrical tank having cross sectional area A is filled with two liquids of density ρ , 2ρ to a height h , $2h$ respectively. A small hole having area a is made in vertical wall so as to achieve maximum range of effluxed liquid on ground. If ($\rho = 1000 \text{ kg/m}^3$, $h = 0.5 \text{ m}$, $A = 1 \text{ m}^2$, $a = 1 \text{ cm}^2$) (Height is taken from ground)



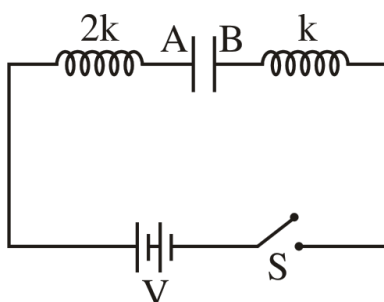
- (A) Height of hole under the condition mentioned is 1m.
(B) Height of hole under the condition mentioned is 0.625 m.
(C) Maximum range of efflux on the ground is 1m.
(D) Maximum range of efflux on the ground is $\sqrt{2}$ m.

Space for Rough Work

4. Nine identical capacitors, each of capacitance $C = 4.2 \text{ mF}$, are connected with a source of emf $E = 2 \text{ V}$ as shown in the Figure below.



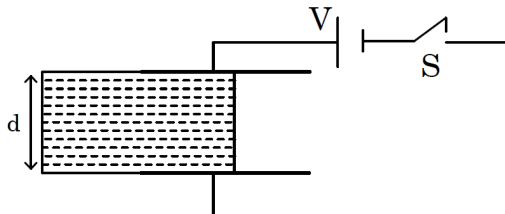
- (A) The equivalent capacitance between A and B is 7 mF
 (B) Charge on capacitor C_7 is 5.2 mC
 (C) The equivalent capacitance between A and B is 11 mF
 (D) None of the above
5. In the given circuit, two identical parallel conducting plates A and B are connected to a 25 V battery by metal springs of spring constant $2k$ and k , respectively. Initially, the switch 'S' is open and the plates are uncharged. In fact, the two plates form a capacitor. When the switch 'S' is closed, distance between the plates becomes 2 mm and it is one third of the initial distance between plates. Also the electric potential energy stored in the capacitor is found to be $7.5 \times 10^{-4} \text{ J}$.



- (A) Spring constant of spring connected to plate A is 281.25 N/m .
 (B) Spring connected to plate B has an extension equal to 2.67 mm .
 (C) Charge on capacitor is equal to $30 \text{ } \mu\text{C}$.
 (D) Total potential energy stored in the system is $1.5 \times 10^{-3} \text{ J}$.

Space for Rough Work

6. A parallel plate capacitor is made up of two square plates, each of area ℓ^2 , separated by a distance d . A battery of emf V volt is connected to the capacitor through a switch S as shown in the Figure. When the switch is closed, the dielectric with proper dimensions enters the capacitor with the initial speed v_0 and fills the entire space between the plates before coming to rest, with velocity uniformly decreasing with distance. Point out incorrect statement/s from the following: [Dielectric constant of the material is K]



- (A) The energy stored in the capacitor decreases linearly with time
- (B) The time rate of change of charge flowing is growing exponentially in time
- (C) The charge of the capacitor increases linearly with the distance covered by the dielectric within the capacitor
- (D) The time rate of change of charge flowing in the circuit decreases linearly with the distance covered by the dielectric within the capacitor

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. In List-II R is a non-zero variable finite resistance for each diagram. Some facts are given in List-I and corresponding figures are given in List-II. Match properly

List I		List II	
(I)	Current passing through 4Ω resistance can be zero	(P)	
(II)	Current passing through 4Ω resistance can be from F to C	(Q)	
(III)	Current passing through 2Ω resistance can be from B to A	(R)	
(IV)	Current through R will be zero	(S)	
		(T)	

- (A) I $\rightarrow$ P,Q,R; II $\rightarrow$ R,S; III $\rightarrow$ T,S; IV $\rightarrow$ P,T
- (B) I $\rightarrow$ P,Q,R,S,T; II $\rightarrow$ P,Q,R; III $\rightarrow$ R,S; IV $\rightarrow$ R,S,T
- (C) I $\rightarrow$ P,Q; II $\rightarrow$ R,S; III $\rightarrow$ P,Q,R,S,T; IV $\rightarrow$ P,R
- (D) I $\rightarrow$ Q; II $\rightarrow$ P,Q,R,S,T; III $\rightarrow$ P,Q,R,S,T; IV $\rightarrow$ S

Space for Rough Work

8. Mark the correct option from List-I and List-II

	List-I		List-II
(I)	A block weighs 15 N in air and 12 N when immersed in water. The specific gravity of the block is	(P)	$2^{5/3}$
(II)	Two identical drops of water are falling through air with a steady velocity 2 m/s. If the drops coalesce, the new velocity will be _____ m/s.	(Q)	12
(III)	A thick shell loses half its weight in water. Relative density of shell is 5.0. What fraction of its volume is hollow ?	(R)	$\frac{3}{5}$
(IV)	A balloon has volume of 10 m^3 . It is filled with hydrogen ($\rho = 0.09 \text{ g/l}$). If the density of air is 1.29 g/l , it can lift a total weight of _____ kg.	(S)	5
		(T)	> 1

(A) I $\rightarrow$ S,T; II $\rightarrow$ R; III $\rightarrow$ P,T; IV $\rightarrow$ P,Q,R,S,T

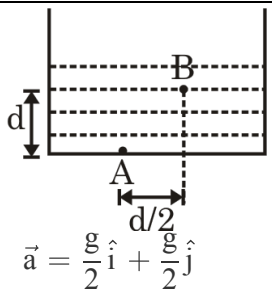
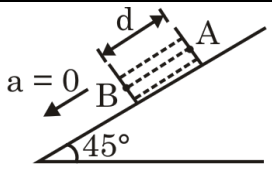
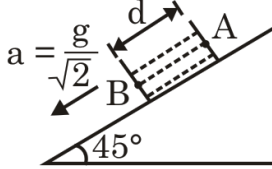
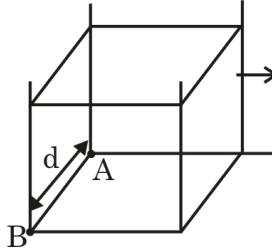
(B) I $\rightarrow$ P,T; II $\rightarrow$ R; III $\rightarrow$ Q,T; IV $\rightarrow$ S,T

(C) I $\rightarrow$ S,T; II $\rightarrow$ P,T; III $\rightarrow$ R; IV $\rightarrow$ P,Q,R,S

(D) I $\rightarrow$ S,T; II $\rightarrow$ P,T; III $\rightarrow$ R; IV $\rightarrow$ P,Q,S,T

Space for Rough Work

9. A vessel filled with liquid of density ρ is accelerating in different situation as shown. List-I gives description of acceleration. List-II gives description of angle made by free surface with horizontal and pressure difference between points A and B as mentioned in List-I.

	List-I		List-II
(I)	 $\vec{a} = \frac{g}{2}\hat{i} + \frac{g}{2}\hat{j}$	(P)	$\tan^{-1}\left(\frac{1}{3}\right)$
(II)	 $a = 0$	(Q)	45°
(III)	 $a = \frac{g}{\sqrt{2}}$	(R)	Zero
(IV)	 $\vec{a} = g\hat{i}$	(S)	$\frac{7}{4}\rho g d$
		(T)	$\frac{\rho g d}{\sqrt{2}}$

(A) I $\rightarrow$ P,S; II $\rightarrow$ R,T; III $\rightarrow$ Q,R; IV $\rightarrow$ Q,R

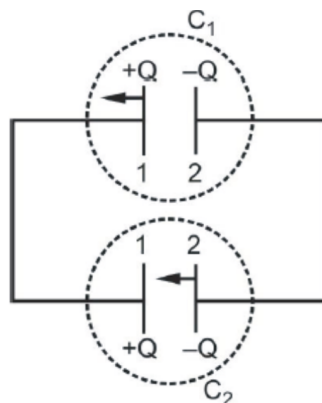
(B) I $\rightarrow$ Q,S; II $\rightarrow$ R,T; III $\rightarrow$ Q; IV $\rightarrow$ Q,R

(C) I $\rightarrow$ P,S; II $\rightarrow$ T; III $\rightarrow$ Q,R; IV $\rightarrow$ Q

(D) I $\rightarrow$ P,S; II $\rightarrow$ R,T; III $\rightarrow$ Q,T; IV $\rightarrow$ Q,R

Space for Rough Work

10. Identical capacitors A and B, each plate area S and separation between the plates d , are connected in parallel. Charge Q is given to each capacitor. Whole arrangement is shown in figure.



Now Keeping one plate of capacitor B fixed another plate is moved towards the first plate with very small constant velocity V from $t = 0$ and keeping one plate of A fixed another plate is moved away from the first plate with small constant velocity V from $t = 0$. Match the proper entries from List-II to List-I for above described system.

	List-I		List-II
(I)	Charge on capacitor A as function of time	(P)	$\frac{(d - Vt)Q}{d}$
(II)	Charge on capacitor B as function of time	(Q)	$\frac{(d + Vt)Q}{d}$
(III)	Current in the circuit as function of time	(R)	Decreases
(IV)	Ratio of electrostatic potential energy Stored between the plates of capacitor A and between the plates of capacitor B as function of time.	(S)	$\frac{QV}{d}$
		(T)	Increase

(A) I $\rightarrow$ P,Q;II $\rightarrow$ R,T;III $\rightarrow$ S;IV $\rightarrow$ R

(B) I $\rightarrow$ P,R;II $\rightarrow$ Q,T;III $\rightarrow$ S;IV $\rightarrow$ R

(C) I $\rightarrow$ P;II $\rightarrow$ Q;III $\rightarrow$ R;IV $\rightarrow$ T

(D) I $\rightarrow$ Q,T;II $\rightarrow$ R,T;III $\rightarrow$ P,S;IV $\rightarrow$ Q,P

Space for Rough Work

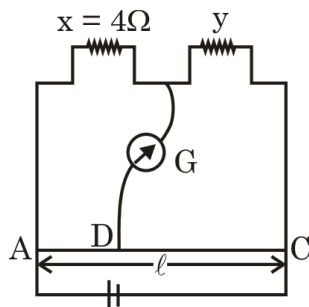
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

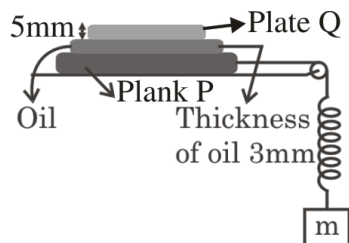
1. The force acting on a particle of mass m is given by $F = mA \frac{e^{-2r}}{r^3}$. The maximum percent error in the measurement of acceleration of the particle at $r = (1 \pm 0.05)\mu\text{m}$ is [A is a constant in proper unit]
2. Figure given here shows a metre bridge wire AC. Which has uniform cross section. The length of wire AC is 100 cm. X is a standard resistor of 4Ω and Y is a coil. When Y is immersed in melting ice, the null point is at 40 cm from point A. When the coil Y is heated to 100°C , a resistor of resistance 12Ω has to be connected in parallel with Y in order to keep the bridge balanced at the same point. Temperature coefficient of resistance of the coil is x . Find value of x .



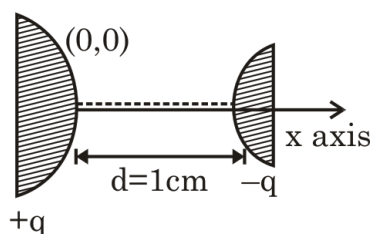
3. A small cube of negligible mass is floating on the surface of lake. The side of the cube is 30 cm and depth of lake is 2m. The cube is pushed very slowly under the water, down to the bottom of the lake. The density of water is 10^3 kg/m^3 and acceleration due to gravity is 9.8 m/s^2 . Find the work done (in Joule) by external pushing force in the process.

Space for Rough Work

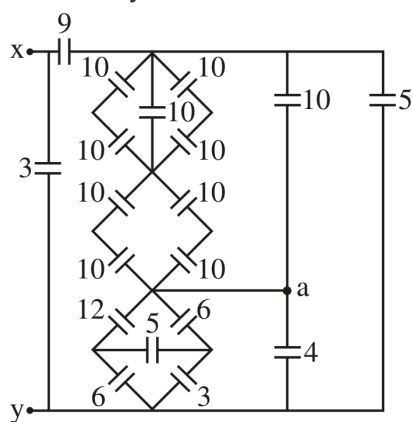
4. A plank P is moved by an external force with velocity of 5 cm/sec. A thin layer of oil of thickness 3 mm separates the moving plank 'P' and metallic plate 'Q' of dimension $10 \text{ cm} \times 20 \text{ cm} \times 5 \text{ mm}$. The plate is in equilibrium and at rest. Coefficient of viscosity of oil is 75 poise and plate is connected to block and spring as shown. $\left(K = 80 \frac{\text{N}}{\text{m}}\right)$, find elongation in spring in cm.



5. There are two conductors A and B of a capacitor which is charged. Due to shape, there is electric field present between the plates $E = \frac{q}{2} (1 + 3 \times 10^4 x^2) \frac{\text{N}}{\text{C}}$, where q is magnitude of charge on capacitor, then capacitance of capacitor in 'F'.

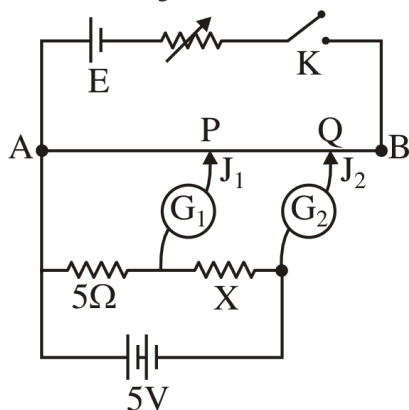


6. Equivalent capacitance (in μF) between x and y is :

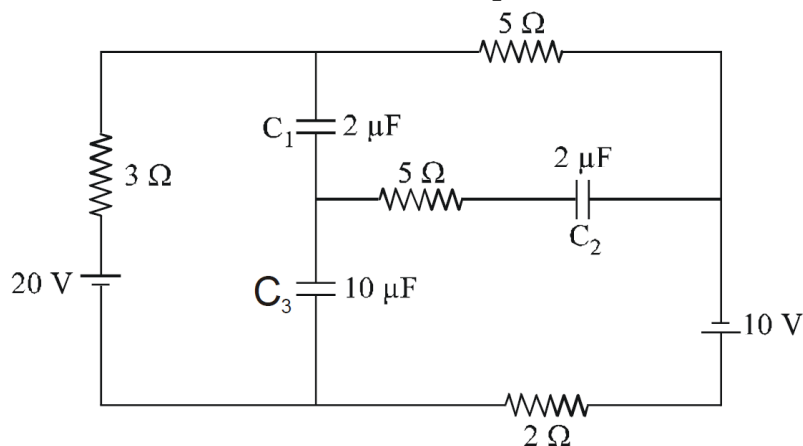


Space for Rough Work

7. A person tries to find the value of unknown resistance using potentiometer as shown in diagram. He uses a resistance of 5Ω , unknown resistance X and a battery of $5V$ in secondary circuit. He touches the jockey J_1 on potential wire to get the point P so that there is no deflection in G_1 , then he locates the point Q , so that G_2 shows zero deflection. It is found that $AP = \frac{AQ}{3}$. Value of X (in Ω) is :-



8. In the circuit shown in figure, find the charge on capacitor C_2 in steady state in μC .



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. An ideal gas whose adiabatic exponent $\left(\frac{C_p}{C_v} = \gamma\right)$ is expanded according to the law $P = \alpha V$, where α is a constant. The initial volume of the gas is equal to V_0 . As a result of expansion, the volume increases 4 times. Select the correct option(s) for above information.

(A) Molar heat capacity of gas in the process is $\frac{R(\gamma + 1)}{2(\gamma - 1)}$

(B) Work done by the gas is $15V_0^2 \frac{\alpha}{2}$

(C) Change in internal energy of gas is $15V_0^2 \frac{\alpha}{2}$

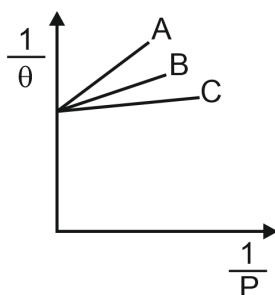
(D) Work done by gas is $\frac{3V_0^2 \alpha}{2}$

Space for Rough Work

2. One mole of gas P and two moles of gas Q are mixed at 27°C and constant pressure to form an ideal gas mixture. Select the correct option(s) based on given information?
- (A) $\Delta S_{\text{mix}} = 3.82 \text{ cal K}^{-1}$ (B) $\Delta H_{\text{mix}} = 0$
 (C) $\Delta S_{\text{mix}} = 0$ (D) $\Delta G_{\text{mix}} = -1145.7 \text{ cal}$
3. A_2O_n is oxidised to AO_3^- by MnO_4^- in acidic medium. If 1.5×10^{-3} mole of A_2O_n requires 60 ml of 0.03 M- KMnO_4 solution for complete oxidation, which of the following statement(s) is/are correct?
- (A) The value of $n = 2$
 (B) Empirical formula of oxide is AO
 (C) 0.15 mole of A_2O_n would require 0.15 mole of acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution for complete oxidation
 (D) 'A' must be a metal from second group of periodic table
4. Which of the following statement is/are correct?
 ['M' represents molar mass of given species]
- (A) $\text{CrI}_3 + \text{KOH} + \text{Cl}_2 \rightarrow \text{K}_2\text{CrO}_4 + \text{KCl} + \text{KIO}_4 + \text{H}_2\text{O}$
 In the balanced chemical reaction, stoichiometric coefficient of KOH is 64
- (B) $(\text{NH}_4)_2 \text{Cr}_2\text{O}_7 \xrightarrow{\Delta} \text{N}_2 + \text{Cr}_2\text{O}_3 + 4\text{H}_2\text{O}$
 Equivalent mass of $(\text{NH}_4)_2 \text{Cr}_2\text{O}_7$ is $(M/12)$
- (C) $[\text{Fe}(\text{CN})_6]^{3-} \rightarrow \text{Fe}^{3+} + \text{CO}_2 + \text{NO}_3^-$
 Equivalent mass for $[\text{Fe}(\text{CN})_6]^{3-}$ is $(M/60)$
- (D) $\text{MnO}_4^{2-} \xrightarrow{\text{H}^+} \text{MnO}_4^- + \text{MnO}_2$
 Equivalent mass of MnO_4^{2-} is $(3M/2)$

Space for Rough Work

5. Which statement(s) is/are TRUE?
- (A) Specific conductance of 0.1 M NaCl is greater than that of 0.2 M NaCl
- (B) Molar conductance of 0.1 M NaCl is greater than that of 0.2 M NaCl
- (C) Equivalent conductance of 0.1 M KOH is greater than its molar conductance
- (D) 0.01 M CH_3COOH offers greater resistance than 0.1 M CH_3COOH in the same conductivity cell
6. Adsorption isotherms of three gases A, B and C are shown in the following figure, where θ is the percentage of surface area of adsorbent (powdered charcoal) covered by the gas and P is the partial pressure of the gas over the adsorbent, then which of the following statement(s) is/are correct? (Consider only monolayer formation during physisorption of gases involved)



- (A) At any finite value of pressure, A shows the highest extent of adsorption
- (B) At any finite value of pressure, C shows the highest extent of adsorption
- (C) Gas A has the highest critical temperature
- (D) Gas C has the highest critical temperature

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7.

List-I Electrode (25° C 1atm)		List-II Oxidation potential (approximate)	
(I)	Pt, H ₂ NaCl(aq)	(P)	0.06 volts
(II)	Pt, H ₂ NH ₄ Cl (K _b = 10 ⁻⁶) (1M)	(Q)	0.6 volts
(III)	Pt, H ₂ HCOOK (K _b = 10 ⁻⁶) (1M)	(R)	0.24 volts
(IV)	Pt, H ₂ KSH (K _{a1} =10 ⁻⁴ , K _{a2} =10 ⁻¹⁰)	(S)	0.42 volts
		(T)	0.36 volt

[Given : $\frac{2.303 RT}{F} = 0.06$]

(A) I → S; II → S; III → P; IV → T

(B) I → S; II → Q; III → R; IV → S

(C) I → S; II → R; III → Q; IV → Q

(D) I → S; II → R; III → Q; IV → S

Space for Rough Work

8.

List-I		List-II	
(I)		(P)	$W_{1 \rightarrow 2 \rightarrow 3} < 0$
(II)		(Q)	$W_{3 \rightarrow 4} > 0$
(III)		(R)	
(IV)		(S)	
		(T)	

(A) I $\rightarrow$ T; II $\rightarrow$ S; III $\rightarrow$ Q; IV $\rightarrow$ S

(B) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ Q; IV $\rightarrow$ S

(C) I $\rightarrow$ P; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ S

(D) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ Q

Space for Rough Work

9. Match the contents of List-I with contents of List-II for 1 mole of gas.

List-I		List-II	
(I)	Raw egg $\rightarrow$ Boiled egg	(P)	$\Delta H = +ve$
(II)	$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$	(Q)	$\Delta S = +ve$
(III)	$H_2O(s) \rightarrow H_2O(l)$ at triple point	(R)	$\Delta G = -ve$
(IV)	Charcoal + Methane $\rightarrow$ adsorbed gas	(S)	$\Delta H = -ve$
		(T)	$\Delta S = -ve$

(A) I $\rightarrow$ P; II $\rightarrow$ R; III $\rightarrow$ R; IV $\rightarrow$ T

(B) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ S

(C) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ R

(D) I $\rightarrow$ T; II $\rightarrow$ Q; III $\rightarrow$ Q; IV $\rightarrow$ T

10. Match the List-I with List-II.

List-I		List-II	
(I)	Molten $PbCl_2$ using inert electrode	(P)	Metal of salt will be reduced
(II)	$NaCl$ aqueous solution using inert electrode	(Q)	$2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-$
(III)	Silver nitrate aqueous solution using Ag electrodes	(R)	Solution becomes basic after electrolysis
(IV)	Sodium nitrate aqueous solution using inert electrodes	(S)	Solution becomes acidic after electrolysis
		(T)	Solution becomes neutral after electrolysis

[Ignore hydrolysis of ions in solution]

(A) I $\rightarrow$ P; II $\rightarrow$ R; III $\rightarrow$ T; IV $\rightarrow$ Q

(B) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ R

(C) I $\rightarrow$ T; II $\rightarrow$ T; III $\rightarrow$ S; IV $\rightarrow$ T

(D) I $\rightarrow$ T; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ T

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

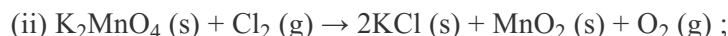
Zero Marks : 0 In all other cases.

1. One litre of 1M H_2O_2 solution is prepared and left open to air. After 10 hours of time, 20ml of this solution could decolourise 20ml of 0.32M KMnO_4 acidified by 0.5M H_2SO_4 . How many grams of O_2 escaped into air from the given solution before titration?
2. Determine the amount of heat given off at constant volume in kcal when 0.5 mole of N_2 and 1.5 mole of H_2 reacted according to equation at 300 K.

$$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g}) \quad \Delta H = -380 \text{ K cal mol}^{-1}$$
3. 2 moles of an ideal gas of unknown heat capacity are in a piston fitted cylinder at 300K. The piston has a mass on the top of it that supplies a constant pressure. The piston is then brought into contact with a thermostat. Heat flows into the system causing the gas to both expand and change temperature. After the temperature has equilibrated, the work done by the gas on the surrounding was found to be 3.32 kJ. Also 11.64 kJ of heat flows into the system. Find value of atomicity of gas. (Take $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$)
4. A molecule A_x dissolves in water and is non volatile. A solution of A_x in water of certain molality showed a depression of 0.93 K in freezing point. The same solution boiled at 100.26°C . When 7.87 g of A_x was dissolved in 100 g of water, the solution boiled at 100.44°C . Given K_f for water = $1.86 \text{ K kg mol}^{-1}$, atomic mass of A = 31 u. Assume no association or dissociation of solute. Calculate the value of x.....

Space for Rough Work

5. Pure oxygen is manufactured by the following sequential reactions, where MnO_2 is acting as a catalyst. The yield in each step of reaction is 50%.



How much mass of KClO_3 in gms is required to liberate total 60 g of O_2 gas in both steps at STP?

(Atomic mass ; K = 39, Cl = 35.5, O = 16)

6. Consider the following statements:

1. Entropy of system increases when an ideal gas undergoes reversible expansion adiabatically.
2. For vaporisation at boiling point, $\Delta G < 0$.
3. For micellisation of surfactants entropy of system increases.
4. Movement of dispersion medium under the influence of an electric field is called cataphoresis.
5. Gold sol is a negatively charged hydrophilic sol.
6. Tyndall effect is more distinct for lyophobic colloid than for lyophilic colloids.
7. Zinc can displace hydrogen from water in standard condition at 298 K.
8. The K_H value of CO_2 is higher than K_H of N_2 at 298 K for dissolution in water.
9. 0.05 M $\text{K}_4[\text{Fe}(\text{CN})_6]$ (50% degree of dissociation) and 0.05 M Mohr's salt (80% degree of dissociation) has same van't hoff factor.
10. On adding 0.1 M KI solution slowly to 0.01 M AgNO_3 solution, causes conductivity to increase gradually and after some time it becomes constant.

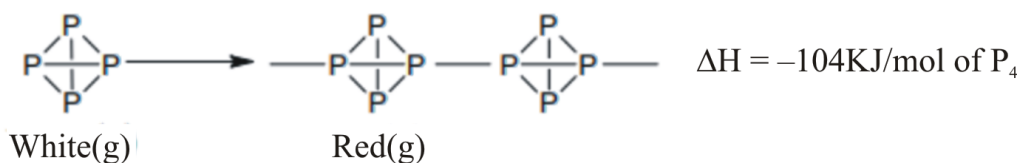
If number of correct statements = x

And number of incorrect statements = y

Then, $\frac{x}{y} =$

Space for Rough Work

7. White phosphorus is a tetra atomic solid $[P_4(s)]$ at room temperature and on strong heating in absence of oxygen, it polymerises into red phosphorus as



The enthalpy of sublimation $[P_4(s) \rightarrow P_4(g)]_{\text{white}}$ is 59 kJ/mol and enthalpy of atomization of $P_4(s)$ is 1265 kJ/mole.

The P – P bond enthalpy in red phosphorus joining the two tetrahedral (in KJ/mol) is

8. Given $\Delta_f H$ values of $CO(g)$ and $CO_2(g)$ are respectively $-30 \text{ kcal mol}^{-1}$ and $-90 \text{ kcal mol}^{-1}$ at 298 K.

The ratio $\left[\frac{\text{heat liberated (Kcal)}}{\text{number of moles of } O_2 \text{ consumed}} \right]$ when one mole of carbon is oxidized and 0.25 mole of carbon forms $CO(g)$ is

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following statements are **INCORRECT**? (where $\{.\}$ denotes the fractional part of function)

(A) $f(x) = \{x\}\cos^2\left(\left(\frac{2x-1}{2}\right)\pi\right)$ is differentiable for all $x \in \mathbb{R}$

(B) $f(x) = |\sin x|\cos^{-1}(\cos x)$ is differentiable $\forall x \in (0, 2\pi)$

(C) $f(x) = ||x-2| - |x-6|| - 3|x| + 2x + 1$ is not differentiable at 3 points

(D) $f(x) = \{x\}|\sin \pi x|$ is not differentiable at all integers

2. If $f(x) = 2|x+1| - |x-1| + 3|x-2| + 2x + 1$ and $h(x) = \begin{cases} 2x & ; x \geq 3 \\ 1 & ; x < 3 \end{cases}$, then choose the correct statements

(A) $y = h(f(x))$ is continuous $\forall x \in \mathbb{R}$

(B) $y = h(f(x))$ is discontinuous at 3 points

(C) $y = h(f(x))$ is not differentiable at 3 points

(D) $y = h(f(x))$ is not differentiable at 4 points

Space for Rough Work

3. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a function satisfying $f(x) e^{f(x)} = x \forall x \in [0, \infty)$, then which of the following statements is(are) TRUE?
- (A) $f(x) \geq 0 \forall x \in (0, \infty)$ (B) $\lim_{x \rightarrow \infty} f(x) = 0$
- (C) $\lim_{x \rightarrow \infty} \frac{f(x)}{\ln x} = 1$ (D) $\lim_{x \rightarrow \infty} \frac{f(x)}{\ln x} = 0$
4. Let $f : (0, \infty) \rightarrow \mathbb{R}$, $f(x) = \left(\frac{x+1}{x}\right)^{\left(x+\frac{1}{2}\right)}$ and $g(x) = \frac{f'(x)}{f(x)}$. Then which of the following statement is/are TRUE ?
- (A) $g(x)$ is increasing in $(0, \infty)$ (B) $\lim_{x \rightarrow \infty} g(x) = 0$
- (C) $\lim_{x \rightarrow \infty} f(x) = e$ (D) $\lim_{x \rightarrow \infty} g(x) = \sqrt{e}$
5. Let $P(x_0, y_0)$ be a point on the curve $C : (x^2 - 11)(y + 1) + 4 = 0$ where $x_0, y_0 \in \mathbb{N}$. If area of the triangle formed by the normal drawn to the curve 'C' at P and the co-ordinate axes is $\left(\frac{a}{b}\right)$, $a, b \in \mathbb{N}$, $[\text{HCF}(a, b) = 1]$
- (A) Equation of normal is $x + 6y = 9$ (B) Area of triangle is $27/4$
- (C) Area of triangle is $25/4$ (D) Equation of normal is $x + 6y = 5$
6. Given two function $F(x) = \left(1 + \frac{1}{x}\right)^x$, $G(x) = \left(1 + \frac{1}{x}\right)^{x+1}$ defined for all $x > 0$. Which of the following is decreasing function for $\forall x > 0$?
- (A) $F(F(x) - G(x))$ (B) $G(F(x) - G(x))$ (C) $G(x) - F(x)$ (D) $G(F(x))$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Let $f(x) = \lim_{n \rightarrow \infty} \prod_{n=0}^{n-1} \frac{(x^{2^n} + 1)^2}{(x^{2^{n+1}} + 1)} \forall x \in [0, \infty) \sim \{1\}$ & $f(1) = 1$, then

	List-I		List-II
(I)	$f(x)$ is increasing in	(P)	$x \in (1, \infty)$
(II)	$f(x)$ is decreasing in	(Q)	$x \in (0, \frac{1}{2}) \cup (5, \infty)$
(III)	$f(x)$ has a local minima in	(R)	$x \in (0, 1)$
(IV)	$f(x) = x$ has only one solution in	(S)	$x \in (0, 2)$
		(T)	$x \in (0, 1) \cup (4, 5)$

(A) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ S

(B) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ P

(C) I $\rightarrow$ S; II $\rightarrow$ T; III $\rightarrow$ R; IV $\rightarrow$ S

(D) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ T; IV $\rightarrow$ R

Space for Rough Work

8. Let $f_1(x) = x \tan^{-1} x$, $f_2(x) = \begin{cases} \frac{x}{\ln 2} & ; \text{ if } x \neq 1 \\ 1 & ; \text{ if } x = 1 \end{cases}$, $f_3(x)$
 and $= \begin{cases} (x+1)e^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)} & ; \text{ if } x \neq 0 \\ 0 & ; \text{ if } x = 0 \end{cases}$ $f_4(x) = \frac{2 \tan\left(\frac{\pi}{4} - x\right)}{\cot 2x}$ if $x \neq \frac{\pi}{4}$

List-I		List-II	
(I)	Number of critical points for $f_1(x)$ over its domain is	(P)	-1
(II)	Derivative of $f_2(x)$ at $x = 1$ is	(Q)	0
(III)	Number of points of discontinuity for $f_3(x)$ in the interval $[-2, 2]$ is	(R)	1
(IV)	Value of $f_4\left(\frac{\pi}{4}\right)$ such that the function is continuous everywhere on interval $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$ is	(S)	2
		(T)	None of these

(A) $I \rightarrow S; II \rightarrow T; III \rightarrow Q; IV \rightarrow Q$

(B) $I \rightarrow R; II \rightarrow T; III \rightarrow P; IV \rightarrow S$

(C) $I \rightarrow R; II \rightarrow T; III \rightarrow R; IV \rightarrow R$

(D) $I \rightarrow Q; II \rightarrow P; III \rightarrow S; IV \rightarrow P$

9. Let $f(x) = \begin{cases} x+a & , x < 0 \\ \sqrt{x^2 - 2x + 1} & , x \geq 0 \end{cases}$
 and $g(x) = \begin{cases} x+1 & , x < 0 \\ x^2 + 2(b-x) + 1 & , x \geq 0 \end{cases}$

a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all real x

List-I		List-II	
(I)	Number of possible values of b is	(P)	0
(II)	Sum of all possible values of a is	(Q)	1
(III)	Value of $\frac{d(g(f(x)))}{dx}$ at $x = 4$ is	(R)	2
(IV)	Product of all possible values of a is	(S)	4
		(T)	greater than 4

(A) $I \rightarrow P; II \rightarrow Q; III \rightarrow S; IV \rightarrow R$

(B) $I \rightarrow P; II \rightarrow Q; III \rightarrow T; IV \rightarrow R$

(C) $I \rightarrow R; II \rightarrow P; III \rightarrow T; IV \rightarrow S$

(D) $I \rightarrow R; II \rightarrow Q; III \rightarrow S; IV \rightarrow P$

Space for Rough Work

10. Let $\lim_{x \rightarrow 0} \left(\frac{a \sin x + bxe^x + 3x^2}{\sin x - 2x + \tan x} \right)$ exists and has value equal to L.

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(I)	The value of 'a' is	(P)	-12
(II)	The value of 'b' is	(Q)	-3
(III)	The value of 'L' is	(R)	3
(IV)	The value of $\frac{b-L}{a}$ is	(S)	6
		(T)	7

(A) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ P; IV $\rightarrow$ S

(B) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ R

(C) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ P

(D) I $\rightarrow$ T; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ T

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let $f: (0, \infty) \rightarrow (0, \infty)$ be a differentiable function such that $f' \left(\frac{a}{x} \right) = \frac{x}{f(x)}$ where a is positive constant and $f(1) = 1$, $f(2) = 2$, then $f(5)$ is equal to
2. Let $f(x)$ be a differentiable function with $f'(0) = 2$ and $f(0) = 3$ such that $f'(x) = \frac{f(x+n) - f(x)}{n} \quad \forall x \in \mathbb{R}$ and $n \in \mathbb{N}$, then $\frac{f(5)}{f(1)}$ is
3. Let $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$ be a polynomial of degree 5 which increases in the interval $(-\infty, 1]$ and $[3, \infty)$ and decreases in the interval $(1, 3)$. Given that $f(0) = 2$, $f(1) = \frac{88}{15}$ and $f'(2) = 0$, then a_5 is equal to

Space for Rough Work

4. $\lim_{t \rightarrow 0} \frac{\sin(t^3) - t^3 \cos(t^3) + t^{15}}{t^3(e^{3t^3} - 1 - 3t^3)}$ is equal to
5. Let $f(x)$ be a real valued function such that $f(x) = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n^2}\right) \left(1 + \frac{2x}{n^2}\right) \dots \left(1 + \frac{nx}{n^2}\right) \forall x \in \mathbb{R}^+$ and $n \in \mathbb{N}$, then $f(\ln 4)$ is
6. If $f(x) = \prod_{n=1}^{50} (x - n)^{n(51-n)}$; then $\frac{f'(51)}{f(51)}$
-

Space for Rough Work

7. If $f(x)$ is differentiable function with $f'(0) = 1$ and $\cos 2x \sin 2y(f(3x + 3y) + f(3x - 3y))$
 $= \sin 2x \cos 2y(f(3x + 3y) - f(3x - 3y))$ for all $x, y \in \mathbb{R}$, then the equation of tangent to $y = f(x)$ at $x = \frac{\pi}{2}$ is
 $2x - ay = \pi - c\sqrt{3}$, then $a + c =$
8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$
 $f(x) = (2x - 3\pi)^3 + \frac{4}{3}x + \cos x$, if $g(x)$ is inverse function of $f(x)$ then value of $g''(2\pi)$
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work